

DATA ANALYSIS BOOTCAMP

Day 1: Concepts, Frameworks & the Real World

ORU

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Today's Roadmap



What Data Analysis Actually Is

Not the textbook version

10 min



How Data Moves Through a Company

Ingestion → Transformation → Dashboard

15 min



The Analysis Framework + Real Examples

Scope → Explore → Analyze → Communicate

25 min



5 Core Analysis Patterns

The techniques that power every project

10 min



Project Kickoff: Toronto Airbnb

Your Day 2 problem statement

10 min

What is Data Analysis?

What people think it is

"Making charts in Excel"

"Running SQL queries"

"Cleaning data all day"

"Building dashboards"

What it actually is

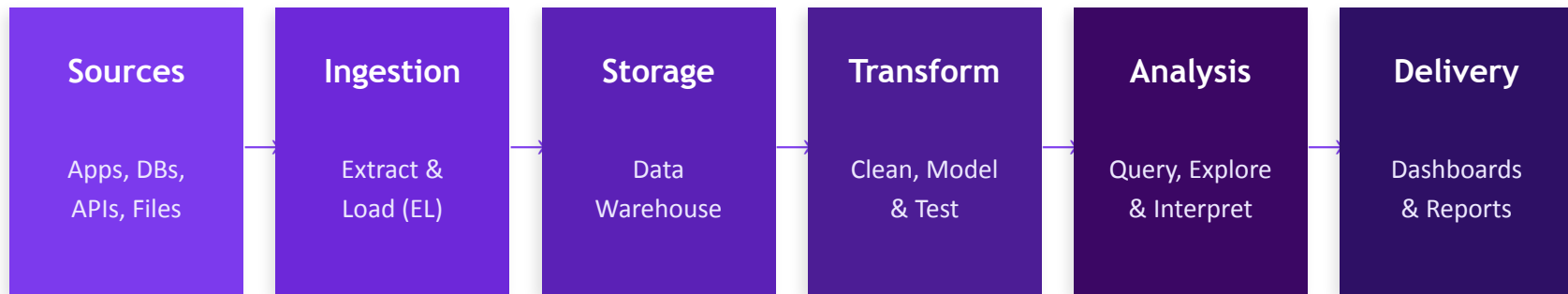
**Turning ambiguous business questions
into clear, evidence-backed decisions.**

The tools are how.

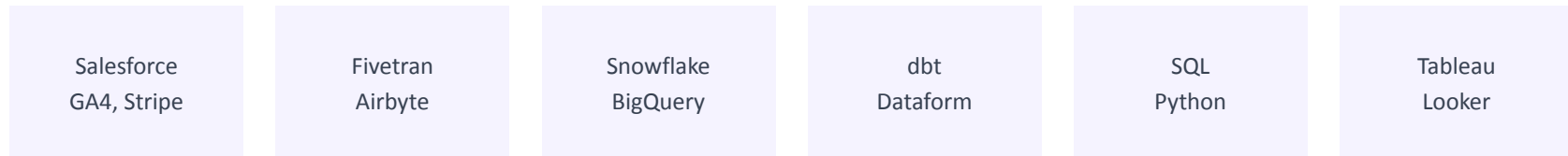
The thinking is what matters.

How Data Moves Through a Company

From raw events to the dashboard someone actually uses



Real-world tools at each stage



Where Every Role Fits

Not just job titles — where they sit in the pipeline and why



Data Engineer

Builds & maintains the pipeline

Ingestion → Storage



Analytics Engineer

Models & transforms data for use

Storage → Transformation



Data Analyst

Explores, interprets, communicates

Analysis → Delivery



Data Scientist

Builds predictive models & experiments

Analysis (deep)

THE ANALYSIS FRAMEWORK

A repeatable system for approaching any data problem

Same 4 steps, every time, regardless of domain or tool

The 4 Stages

01



SCOPE

Define the question.
What decision does
this inform?

02



EXPLORE

Profile the data.
What do you have?
What's missing?

03



ANALYZE

Segment, compare,
find patterns.
Test hypotheses.

04



DELIVER

Structure findings.
Lead with the
so-what.

Framework in Action: E-Commerce

Scenario: Cart abandonment rate jumped from 65% to 78% in one month. CEO asks: "What happened?"

SCOPE

What decision does this inform? → Should we fix checkout UX, adjust pricing, or investigate a technical bug?
Who's the audience? → VP Product + VP Eng. They need a root cause, not a dashboard.

EXPLORE

Pull funnel data by step, device, user type. First pass: mobile abandonment is 89% vs desktop 64%. Payment step has the biggest drop-off. New users 3x more likely to abandon than returning users.

ANALYZE

Segment by payment method: credit card completion is fine, but a new "buy now pay later" option added last month errors on mobile 40% of the time. That one feature broke the mobile funnel.

DELIVER

Lead: Mobile BNPL integration is broken and caused the spike. Evidence: 2 charts (funnel by device, error rate by payment method). Recommendation: Disable mobile BNPL, fix, re-launch. Projected recovery: 10-12% cart rate.

Same 4 steps. The framework doesn't change — only the domain does.

Framework in Action: SaaS

Scenario: Monthly churn doubled from 3% to 6%. CS team blames pricing. Product team blames onboarding. Who's right?

SCOPE

Decision: Where should engineering invest — pricing flexibility, onboarding flow, or feature adoption? Audience: CEO + Head of Product. They need the root cause, not a churn number.

EXPLORE

Pull churn cohorts by signup month, plan tier, and feature usage. 70% of churned users never completed onboarding. Only 12% cited pricing in exit surveys. Median churner used 1 of 8 core features.

ANALYZE

Users who completed onboarding AND used 3+ features had 2% churn. Users who didn't complete onboarding had 14% churn. Pricing tier made almost no difference once you controlled for activation.

DELIVER

Lead: Churn is an activation problem, not a pricing problem. 70% of churners never onboarded. Rec: Redesign first-week experience to drive 3-feature activation. Pricing changes won't move the needle.

Two teams had opinions. The data pointed somewhere neither expected.

Same Framework, Every Domain

The stages don't change. The vocabulary does.

Stage	Retail	Healthcare	Finance
SCOPE	Why are returns up 20% this quarter?	Why are ER wait times above target?	Why did loan defaults spike in Q3?
EXPLORE	Pull returns by category, channel, return reason	Pull wait times by shift, triage level, staffing ratio	Pull defaults by loan type, credit band, origination month
ANALYZE	Online apparel returns are 3x store. Size guides are outdated for 4 categories.	Night shifts have 40% fewer nurses per patient. Wait times correlate with staffing, not volume.	Auto loans originated with new underwriting model have 2x default rate. Model issue.
DELIVER	Update size guides for 4 categories. Projected 15% return rate reduction.	Add 2 nurses to night shift. Projected 25 min wait time reduction.	Pause new model, revert to previous scoring. Review model training data for bias.

Notice: the thinking is identical. Retail returns, ER waits, loan defaults — same structure.

01 Scope the Problem

Before touching any data, answer these:

What decision will this analysis inform?

If nobody will act on it, don't build it.

Who is the audience?

A CFO and a product manager need different things.

What does 'good enough' look like?

Perfect is the enemy of shipped.

What data do I actually have access to?

Constraints shape the approach. Work with what exists.

02 Explore & Profile

Understand the shape of your data before analyzing it

Data Shape

- How many rows and columns?
- What data types are present?
- What's the grain? (one row = what?)
- Any nulls, duplicates, outliers?

Data Quality

- What's the time range?
- What are the categorical breakdowns?
- Are there joins or lookups needed?
- What can you NOT answer with this data?

03 Analyze

This is where you earn your keep

Segmentation

Break data into meaningful groups. By geography, type, time period, user behavior.

Comparison

Compare segments against each other or a baseline. Where are the differences?

Correlation

When X changes, does Y change? Is it causal or coincidental?

Distribution

How is the data spread? Normal, skewed, bimodal? Where are the outliers?

04 Deliver

The best analysis is worthless if nobody understands it

The Delivery Structure

1

Lead with the answer

Don't make them sit through your methodology to get the punchline.

2

Show the evidence

The 2-3 charts or metrics that support your conclusion. No more.

3

Acknowledge the gaps

What you couldn't answer. What would need more data. This builds trust.

4

End with a recommendation

Not 'here's the data' but 'here's what I'd do.' Analysts who recommend get promoted.

5 Core Analysis Patterns

Every analysis you'll ever do uses one or more of these

Trend

How does X change over time?

Monthly churn rate over 12 months. Is it getting better or worse?

Line chart

Composition

What makes up the whole?

Revenue by product line. What % comes from each source?

Stacked bar / Pie

Distribution

How is X spread?

Listing counts across 140 neighbourhoods. Concentrated or dispersed?

Histogram / Box plot

Comparison

How does A differ from B?

Professional vs casual host performance. Who gets more reviews?

Grouped bar / Table

Relationship

When X changes, what happens to Y?

Does higher availability correlate with more reviews per month?

Scatter plot

Bad Analysis vs Good Analysis



Describes the data

"Average order value is \$47."

"We have 15,776 Airbnb listings in Toronto."

"67% of listings are entire homes."

These are facts. They don't answer anything.

The reader's reaction: "OK... so what?"



Answers a question

"Mobile AOV (\$32) is 40% lower than desktop (\$53). Mobile users add fewer items. Rec: add cross-sell widget."

"Waterfront has 17% of supply but review velocity is below average — the market is saturated. Target Kensington-Chinatown instead: half the listings, 2x the demand signal."

Comparison + segmentation + recommendation.

The reader knows what to do next.

YOUR PROJECT

Toronto Airbnb Market Intelligence

Real data. Real questions. Real deliverable.

Inside Airbnb · 15,776 listings · 140 neighbourhoods

The Brief

You're a data analyst at a property management startup expanding into Toronto's short-term rental market. **Leadership needs a market intelligence report before committing resources.**

Your job: Analyze the Toronto Airbnb market to identify where supply is concentrated, which areas show the strongest demand signals, what the competitive landscape looks like, and how regulatory compliance varies. Deliver findings as a Tableau dashboard.

Questions to Answer

1. Where is the supply concentrated? Which neighbourhoods and room types dominate?
2. Where is demand strongest? (Using review activity as a proxy for bookings)
3. Who are we competing with? Professional operators vs casual hosts?
4. How does regulatory compliance (licensing) vary across the city?

The Dataset

Inside Airbnb · Toronto · listings.csv

15,776

Listings

140

Neighbourhoods

4

Room Types

12,545

With Reviews

Column	What It Tells You
neighbourhood	Where the listing is located (140 areas)
room_type	Entire home, Private room, Shared, Hotel
minimum_nights	Short-term (1-7) vs long-term (28+) strategy
number_of_reviews	Total lifetime review count (demand proxy)
reviews_per_month	Review velocity (recent demand signal)
availability_365	Days available in next year (supply signal)
calculated_host_listings_count	How many listings the host operates
license	Regulatory compliance indicator

Day 2 Preview

What you'll build tomorrow

1

Load & Profile

Get the data into Python (Colab) or BigQuery. Run initial profiling.

Python / SQL

20 min

2

Clean & Transform

Handle nulls, categorize minimum_nights, flag professional hosts.

Python / SQL

20 min

3

Analyze

Answer the 4 questions. Segment, compare, find patterns.

Python / SQL

25 min

4

Dashboard

Build a 3-sheet Tableau dashboard with your findings.

Tableau Public

25 min

Before Day 2

1

Download the dataset

listings.csv from the GitHub repo (link provided)

2

Set up your environment

Option A: Google Colab (just a browser)

Option B: BigQuery Sandbox (free, no credit card)

3

Install Tableau Public

Free at public.tableau.com — works on Mac and Windows

4

Write down 3 hypotheses

What do you THINK is true about Toronto's Airbnb market?

We'll test these tomorrow.

See you on Day 2.

Come with your hypotheses ready.

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